

Operations Research & Applications programming

Ch. 3

Artificial Variable Techniques –

Big M-method

Introduction

- ❑ LPP in which all the constraints are $(\leq)$ with nonnegative right-hand sides offer a convenient all-slack starting basic feasible solution. Models involving $(=)$ and/or $(\geq)$ constraints do not.
- ❑ The procedure for starting "ill-behaved" LPP with $(=)$ and $(\geq)$ constraints is to use **artificial variables** that play the role of slacks at the first iteration, and then dispose of them legitimately at a later iteration.
- ❑ Two closely related methods are introduced here: the Big *M-method* and the two-phase method.
- ❑ *In this lecture*, we discuss the Big M method.

Artificial Variable Techniques – Big M-method

- Suppose that a constraint equation i does not have a slack variable. i.e. there is no i^{th} unit vector column in the LHS of the constraint equations. (This happens for example when the i^{th} constraint in the original LPP is either $\geq$ or $=$) Then we augment the equation with an artificial variable R_i to form the i^{th} unit vector column.

Artificial Variable Techniques – Big M-method

- However, because the artificial variables are not part of the original LP model, they are assigned a very high **penalty** in the objective function, thus forcing them (eventually) to equal zero in the optimum solution. This is achieved by giving a large **penalty** to the coefficient of the artificial variable in the objective function as follows:
 - = - M in a *maximization* problem,
 - = M in a *minimization* problemwhere M is a very large positive number.

Artificial Variable Techniques – Big M-method

□ *Example* : Consider the LPP

$$\text{Minimize } z = 2x_1 + x_2$$

□ Subject to the constraints :

$$3x_1 + x_2 \geq 9$$

$$x_1 + x_2 \geq 6$$

$$x_1, x_2 \geq 0$$

□ Putting this in the standard form, the LPP is:

Artificial Variable Techniques – Big M-method

$$\text{Minimize } z = 2x_1 + x_2$$

□ Subject to the constraint :

$$3x_1 + x_2 - s_1 = 9$$

$$x_1 + x_2 - s_2 = 6$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Here s_1 and s_2 are *surplus* variables.

□ *Note* that we do NOT have a 2x2 identity submatrix in the LHS.

Artificial Variable Techniques – Big M-method

- ❑ Introducing the artificial variables, R_1, R_2 , the LPP is modified as follows:

$$\text{Minimize } z = 2x_1 + x_2 + 0s_1 + 0s_2 + M R_1 + M R_2$$

- ❑ Subject to the constraints

$$3x_1 + x_2 - s_1 + R_1 = 9$$

$$x_1 + x_2 - s_2 + R_2 = 6$$

$$x_1, x_2, s_1, s_2, R_1, R_2 \geq 0$$

- ❑ *Note* that we now have a 2x2 identity submatrix in the coefficient matrix of the constraint equations.

Artificial Variable Techniques – Big M-method

- Now we solve the above LPP by the Simplex method.
- From the standpoint of solving the problem on the computer, M must assume a numeric value. In the manually solution M manipulated algebraically in all the simplex tableaus.
- ($\text{New } z\text{-row} = \text{Old } z\text{-row} + (M.R_1\text{-row} + M.R_2\text{-row})$) **min**
- ($\text{New } z\text{-row} = \text{Old } z\text{-row} - (M.R_1\text{-row} + M.R_2\text{-row})$) **max**

Artificial Variable Techniques – Big M-method

Basic	x1	x2	s1	s2	R1	R2	Sol.
Z	$-2+4M$ -2	$-1+2M$ -1	$-M$ 0	$-M$ 0	0 -M	0 -M	$15M$ 0
R1	3	1	-1	0	1	0	9
R2	1	1	0	-1	0	1	6
Z	0	$-1/3+2M/3$	$-2/3 + M/3$	$-M$	$2/3-4M/3$	0	$6+3M$
x1	1	1/3	-1/3	0	1/3	0	3
R2	0	2/3	1/3	-1	-1/3	1	3
Z	0	0	-1/2	-1/2	$1/2-M/2$	$1/2-M/2$	15/2
x1	1	0	-1/2	1/2	1/2	-1/2	3/2
x2	0	1	1/2	-3/2	-1/2	3/2	9/2

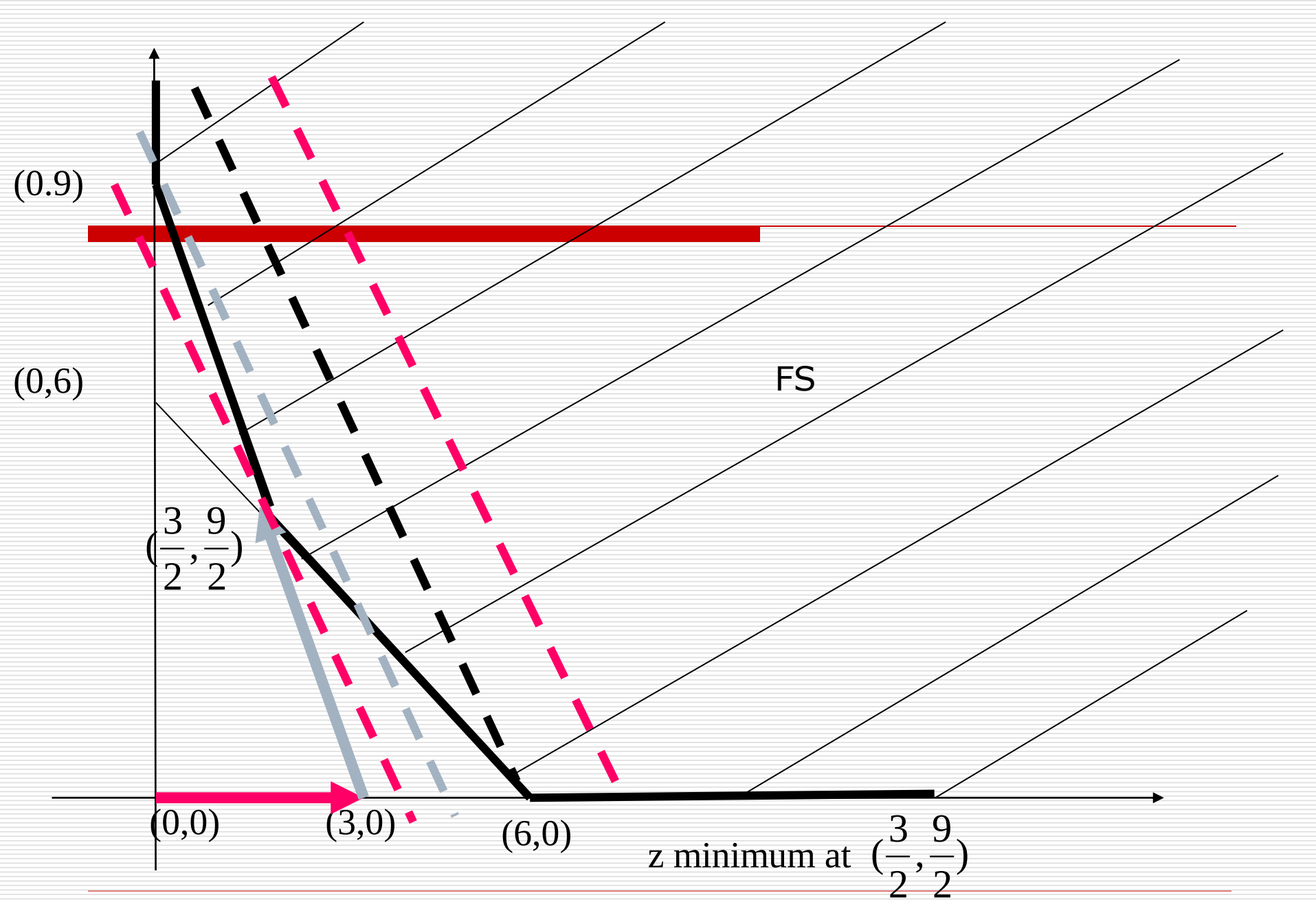
Artificial Variable Techniques – Big M-method

- Note that we have got the optimal solution to the given problem as

$$x_1 = \frac{3}{2}, \quad x_2 = \frac{9}{2}$$

- Optimal $z = \text{Minimum } z = \frac{15}{2}$

- It is illuminating to look at the graphical solution also.



Artificial Variable Techniques – Big M-method

□ *Example* : Consider the LPP

$$\text{Maximum } z = 2x_1 + 3x_2 - 5x_3$$

Subject to the constraints :

$$x_1 + x_2 + x_3 = 7$$

$$2x_1 - 5x_2 + x_3 \geq 10$$

$$x_1, x_2, x_3 \geq 0$$

□ Introducing surplus and artificial variables, s_2 , R_1 and R_2 , the LPP is modified as follows:

Artificial Variable Techniques – Big M-method

$$\text{Maximum } z = 2x_1 + 3x_2 - 5x_3 + 0s_2 - MR_1 - MR_2$$

□ Subject to the constraint :

$$x_1 + x_2 + x_3 + R_1 = 7$$

$$2x_1 - 5x_2 + x_3 - s_2 + R_2 = 10$$

$$x_1, x_2, x_3, s_2, R_1, R_2 \geq 0$$

□ Now we solve the above LPP by the Simplex method.

Artificial Variable Techniques – Big M-method

Basic	x1 	x2	x3	s2	R1	R2	Sol.
Z	-2-3M -2	-3+4M -3	5-2M 5	M 0	0 +M	0 +M	-17M 0
R1	1	1	1	0	1	0	7
R2	2	-5	1	-1	0	1	10
Z	0	 -8-7M/2	6 -M/2	-1-M/2	0	1+3M/2	10-2M
R1	0	7/2	1/2	1/2	1	-1/2	2
x1	1	-5/2	1/2	-1/2	0	1/2	5
Z	0	0	50/7	1/7	16/7+M	-1/7+M	102/7
x2	0	1	1/7	1/7	2/7	-1/7	4/7
x1	1	0	6/7	-1/7	5/7	1/7	45/7

Artificial Variable Techniques – Big M-method

- The Optimum (Maximum) value of $z = 102/7$
and it occurs at :

$$x_1 = 45/7, x_2 = 4/7, x_3 = 0$$

Remarks

- If in any iteration, there is a tie for entering variable between an artificial variable and other variable (decision, surplus or slack), we must prefer the non-artificial variable to enter the basis.
- If in any iteration, there is a tie for leaving variable between an artificial variable and other variable (decision, surplus or slack), we must prefer the artificial variable to leave the basis.

Remarks

- If in the final optimal tableau, an artificial variable is present in the basis at a non-zero level, this means our original problem has *no feasible solution*.

Artificial Variable Techniques – Big M-method

- ❑ Consider the LPP

$$\text{Minimize } z = 4x_1 + 6x_2$$

- ❑ Subject to the constraints :

$$-2x_1 + 3x_2 = 3$$

$$4x_1 + 5x_2 \geq 10$$

$$4x_1 + 8x_2 \geq 5$$

$$x_1, x_2 \geq 0$$

Artificial Variable Techniques – Big M-method

- ❑ Introducing the artificial variables, R_1, R_2, R_3 the LPP is modified as follows:

$$\text{Minimum } z = 4x_1 + 6x_2 + MR_1 + MR_2 + MR_3$$

- ❑ Subject to the constraints

$$-2x_1 + 3x_2 + R_1 = 3$$

$$4x_1 + 5x_2 - s_2 + R_2 = 10$$

$$4x_1 + 8x_2 - s_3 + R_3 = 5$$

$$x_1, x_2, s_2, s_3, R_1, R_2, R_3 \geq 0$$

Artificial Variable Techniques – Big M-method

Basic	x1	x2 	s2	s3	R1	R2	R3	Sol.
z	-4+6M -4	-6+16M -6	-M 0	-M 0	0 -M	0 -M	0 -M	18M 0
R1	-2	3	0	0	1	0	0	3
R2	4	5	-1	0	0	1	0	10
 R3	4	8	0	-1	0	0	1	5
z	-1-2M	0	-M	-3/4+M 	0	0	3/4-2M	5/4+8M
 R1	-7/2	0	0	3/8	1	0	-3/8	9/8
R2	3/2	0	-1	5/8	0	1	-5/8	55/8
x2	1/2	1	0	-1/8	0	0	1/8	5/8

Artificial Variable Techniques – Big M-method

Basic	x1	x2	s2	s3	R1	R2	R3	Sol.
z	-8 +22/3M	0	-M	0	2+ 8/3M	0	-M	6+5M
s3	-28/3	0	0	1	8/3	0	-1	3
R2	22/3	0	-1	0	-5/3	1	0	5
x2	-2/3	1	0	0	1/3	0	0	1
z	0	0	-6/11	0	2/11- M	12/11-M	-M	126/11
s3	0	0	-14/11	1	6/11	14/11	-1	103/11
x1	1	0	-3/22	0	-5/22	3/22	0	15/11
x2	0	1	-1/11	0	2/11	1/11	0	16/11

Artificial Variable Techniques – Big M-method

□ This is the optimal tableau.

The optimal solution is :

$$x_1 = 15/22, x_2 = 16/11, x_3 = 0$$

And the optimum $z = \text{Min } z = 126/11$